

**PART A: PLANNING**

A rock climber finds that his favourite rope has a few knots in it that he can't undo. He wants to make sure that his rope is still safe to use. He has asked you to test out if knots affect how much weight a rope can hold without breaking.

1. Draw a labelled diagram to show how you would conduct this experiment. (3)

- must be - 2D, scientific
- labelled.
- captioned (ie Fig 1: Apparatus for experiment).
or title will do for yr 9 - but explain caption.

eg

The diagram shows a horizontal support line with two vertical lines hanging from it. Each vertical line has a small circle (representing a knot) and a square weight at the bottom.

2. Name the independent variable (1)

number of knots (or type).

3. Name the dependent variable (1)

weight / force before breaking.

4. Name 3 controlled variables. (3) each

eg. type of rope width / thickness of rope
type of knot how weight is applied.
length of rope etc.

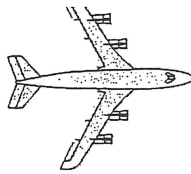
5. Write a prediction or hypothesis for this experiment (1)

eg Increasing the number of knots will increase / decrease / not affect breaking strength.
must state relationship (ie not "will affect")
must be specific and measurable, ie not "less safe"
No "because"...

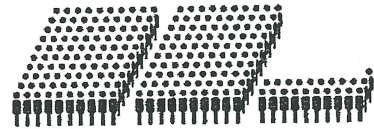
6. List two safety precautions that you need to consider for this experiment. (2) (each)

eg not dropping weights on head / foot
working at safe heights safety harness
not lifting heavy weights
head protection closed in shoes.

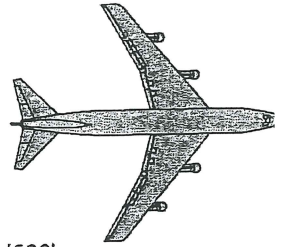
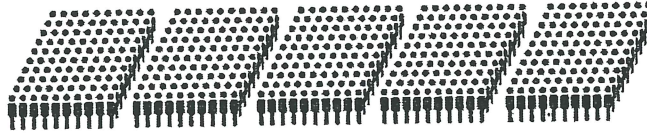
PART B: RECORDING DATA



Boeing 707 (221)



Car (4)



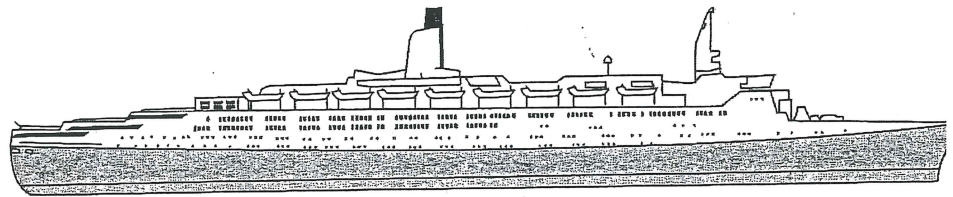
Boeing 747 (500)



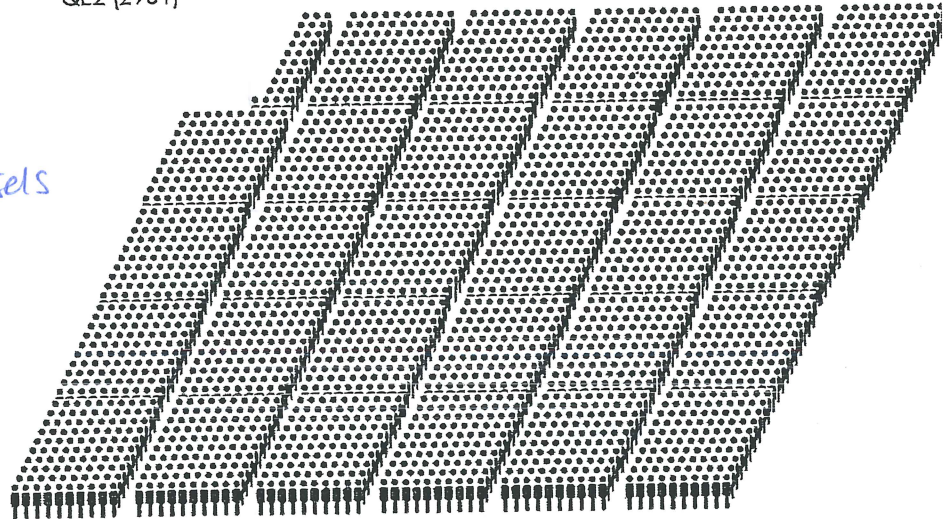
Single decker bus (43)



Double decker bus (72)



QE2 (2931)



Passengers carried by diff. vessels

Vessel	Passengers
Car	4
Bus	43
Double decker	72
Boeing 707	221
747	500
QE2	2931

7. Make a table for the information shown in the above diagram, which shows the amount of passengers that each type of vessel can carry (4)

Table Must Have:

- Title that relates variables (eg Amount of passengers different vessels can carry).
- Data in vertical columns
- headings for each column.
- data accurately recorded.

IV	DV

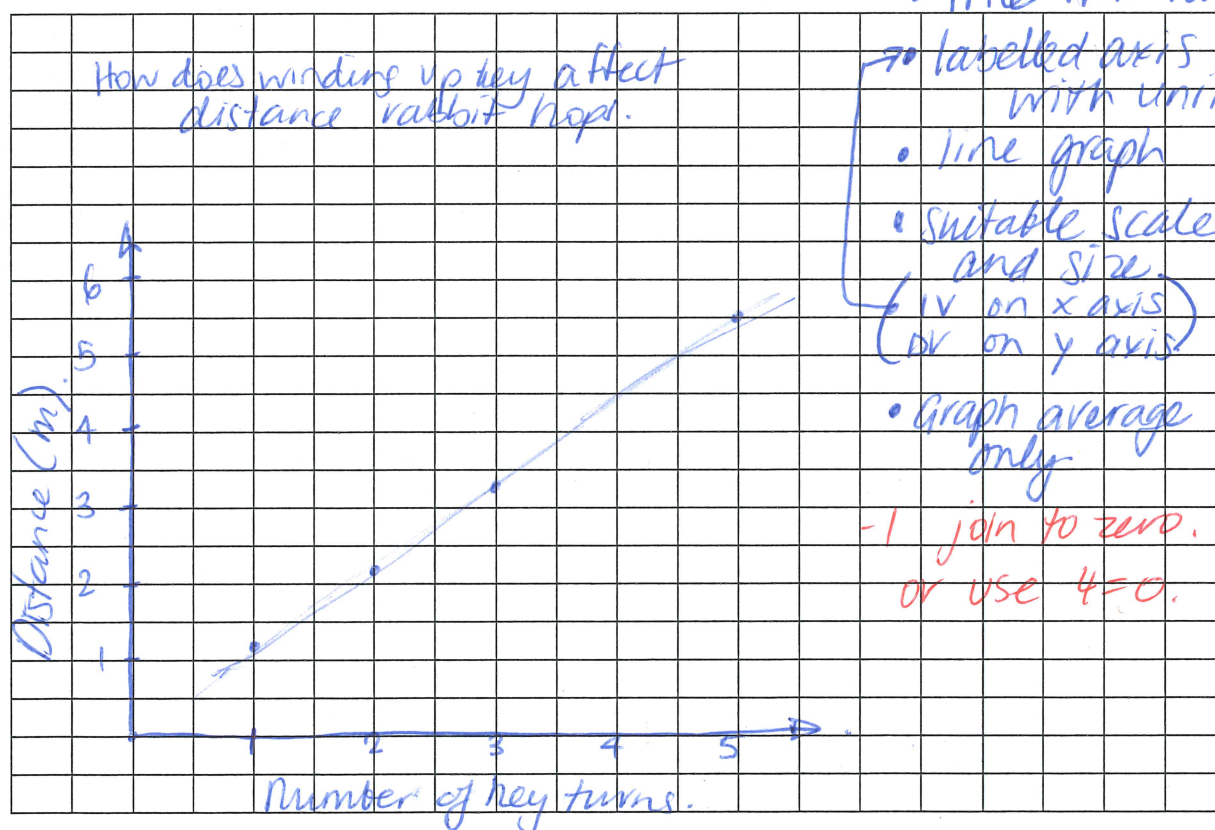
ANALYSING DATA

A student just bought a new windup toy rabbit. The rabbit's legs are powered by springs that are wound up with a key. The student wanted to see if turning the windup key would make the rabbit hop further. Here are the results:

How does winding up the key affect the distance the rabbit hopped?

Number of turns of the key	Distance the toy hopped in meters			Average
	Trial 1	Trial 2	Trial 3	
1	1.0	1.3	1.3	1.2
2	2.0	2.4	2.2	2.2
3	3.0	3.6	3.3	3.3
4	No result recorded			
5	5.0	5.9	5.6	5.5

8. Graph this information to show the trends in the results. (5)



9. How far would the rabbit hop if you turned the key four times? (1)

~ 4.4 (4.3 $\rightarrow$ 4.5 acceptable answers).

10. Write a conclusion from these results? (1)

The more turns of the key, the further the rabbit hops.

for next year

- marks for calculating average.

PART C:

Include hypothesis (for conclusion
to comment on).